

TUNGOKOCHEN, Russian Federation

WMO: 306640

Lat: 53.533N

Lon: 115.617E

Elev: 2662

StdP: 13.34

Time Zone: 9.00 (E09)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.7	-42.2	-49.7	0.3	-45.0	-46.6	0.4	-42.0	14.1	-1.7	12.8	2.1	0.2	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	81.3	62.3	78.0	61.7	74.9	60.5	65.6	74.8	63.9	73.4	62.4	71.6	5.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	94.0	69.1	60.7	87.6	67.3	58.9	82.0	65.9	32.0	74.6	30.7	73.4	29.5	71.5	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.3	12.4	DB	-50.8	86.5	2.8	4.6	-52.8	89.8	-54.4	92.5	-56.0	95.0	-58.0	98.3	(4)
(5)			WB	-50.1	68.9	2.6	1.9	-52.0	70.2	-53.5	71.3	-55.0	72.4	-56.9	73.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	-19.6	-10.4	7.4	28.9	43.7	57.2	61.5	57.1	43.7	25.9	-1.4	-18.5	(6)	
	DBStd	30.63	10.08	10.02	12.83	9.34	6.81	6.12	4.45	5.98	7.62	9.51	13.79	11.01	(7)	
	HDD50	10678	2156	1690	1320	635	218	13	1	14	218	748	1543	2122	(8)	
	HDD65	15309	2621	2110	1785	1084	662	241	124	249	640	1213	1993	2587	(9)	
	CDD50	871	0	0	0	1	22	229	357	234	28	0	0	0	(10)	
	CDD65	28	0	0	0	0	0	8	16	4	0	0	0	0	(11)	
	CDH74	834	0	0	0	1	8	293	397	121	14	0	0	0	(12)	
	CDH80	145	0	0	0	0	0	68	67	9	1	0	0	0	(13)	
(14) Wind	WSAvg	3.8	1.7	2.5	4.5	6.0	6.2	4.4	3.7	3.5	4.5	4.2	2.5	1.4	(14)	
(15) Precipitation	PrecAvg	15.6	0.2	0.2	0.2	0.6	1.4	2.6	3.7	3.3	2.1	0.6	0.4	0.2	(15)	
	PrecMax	19.3	0.6	0.6	0.6	1.6	3.0	5.1	6.7	6.2	4.6	1.7	1.5	0.7	(16)	
	PrecMin	9.2	0.0	0.0	0.0	0.1	0.0	0.4	1.2	2.0	0.3	0.0	0.0	0.0	(17)	
	PrecStd	2.4	0.1	0.1	0.2	0.4	0.7	1.2	1.3	0.9	1.1	0.4	0.3	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	28.3	47.5	65.0	75.5	87.0	85.5	81.1	75.8	57.8	39.1	23.3	(19)	
		MCWB	12.1	23.6	36.6	47.5	55.6	65.4	63.9	62.4	57.1	45.2	32.8	20.0	(20)	
	2%	DB	9.0	20.3	41.3	56.4	70.1	81.2	81.7	77.3	68.9	51.5	30.4	11.1	(21)	
		MCWB	6.8	16.6	32.8	42.2	52.4	61.2	63.3	61.8	54.3	40.9	26.2	9.3	(22)	
	5%	DB	3.5	15.4	35.8	50.9	66.0	77.1	78.5	73.7	63.5	46.0	25.8	4.1	(23)	
		MCWB	2.0	12.6	28.9	39.2	49.8	60.1	62.5	60.9	51.7	37.2	22.7	3.0	(24)	
	10%	DB	-1.5	9.6	30.1	46.0	61.1	73.1	75.5	70.1	58.8	41.2	20.5	-1.2	(25)	
		MCWB	-2.6	7.6	25.0	36.2	47.0	58.6	62.0	59.7	50.1	34.3	18.2	-2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	23.0	37.2	48.1	56.8	67.4	68.2	66.6	59.6	45.5	32.4	20.3	(27)	
		MCDB	15.1	27.8	47.0	62.5	74.2	80.9	76.8	73.9	69.1	56.9	38.9	23.7	(28)	
	2%	WB	6.5	16.6	32.8	43.0	53.3	63.8	66.2	64.4	56.8	40.9	26.8	9.5	(29)	
		MCDB	8.3	20.1	41.1	55.4	68.4	75.6	75.3	72.4	64.5	50.5	29.9	11.3	(30)	
	5%	WB	1.4	12.6	28.9	39.7	50.6	61.9	64.6	62.6	54.1	37.8	22.5	3.1	(31)	
		MCDB	2.8	15.2	35.5	50.2	63.7	73.1	74.2	70.1	61.4	45.5	25.7	4.2	(32)	
	10%	WB	-3.4	7.6	25.3	36.5	47.9	60.0	63.2	60.9	50.7	34.6	18.0	-1.9	(33)	
		MCDB	-2.4	9.5	29.9	45.3	59.8	70.7	72.7	67.9	58.8	40.7	20.2	-1.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	26.4	35.3	35.5	25.5	25.7	26.3	24.8	23.5	24.6	24.3	26.4	24.1	(35)	
		MCDBR	30.1	36.5	35.9	34.3	36.7	35.5	33.2	31.1	33.5	30.5	26.8	27.4	(36)	
	5% WB	MCWBR	28.8	33.0	27.8	21.9	20.5	17.6	16.6	16.5	19.5	20.4	22.5	25.7	(37)	
		MCWBR	30.3	36.1	34.6	33.1	33.3	29.6	26.7	24.2	28.2	29.4	26.0	27.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.241	0.271	0.304	0.394	0.506	0.477	0.424	0.386	0.334	0.292	0.263	0.243	(40)	
		taud	2.194	2.164	2.163	2.073	1.839	2.031	2.249	2.342	2.439	2.443	2.273	2.149	(41)	
	Ebn at Noon		235	263	277	262	237	247	260	264	267	257	224	206	(42)	
		Edh at Noon	20	29	36	46	61	51	40	35	28	22	18	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		397	756	1296	1654	1728	1868	1728	1468	1125	752	430	298	(44)	
	RadStd		13	15	29	99	118	126	151	82	92	49	25	10	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page